Master's Thesis

Enhancement of Polyamide Composite Properties Using Nano-Copper Oxide Powder

Abstract

Polyamide (PA) occupies a significant position in engineering plastics due to its excellent mechanical properties, wear resistance, and processability. However, the deficiencies of PA in friction and wear, antistatic behavior, and functional characteristics limit its application in specific scenarios. Nano-copper oxide (CuO), as a low-cost and readily available functional filler with unique physicochemical properties, provides a new pathway for enhancing the performance of PA composites. This thesis systematically investigates the effects of nano-CuO on polyamide 6 (PA6) and polyamide 66 (PA66) properties, covering preparation methods, dispersion strategies, mechanical properties, tribological performance, and functional characteristics. Results demonstrate that within the CuO content range of 0.3–0.5 wt%, PA6 composite tensile strength increases by 47.5%, elongation at break increases by 62.4%, and impact toughness increases by 170.5%. In the PA66 system, CuO and gamma irradiation synergistically enhance flexural strength and flexural modulus. Nano-CuO@$BaSO_4$ co-precipitate filler achieves 100% bactericidal rate at only 0.24 wt% addition while simultaneously improving mechanical properties and thermal stability. This thesis provides systematic theoretical foundations and experimental data for the rational design of functional PA/CuO composites.

Keywords: Polyamide 6; Polyamide 66; Nano-copper oxide; In-situ polymerization; Mechanical properties; Tribological performance

Chapter 1 Introduction

1.1 Research Background

Polyamide (PA), commonly known as nylon, is a class of engineering thermoplastics whose main chain contains amide groups (-NH-CO-). PA6 and PA66 are the two most important varieties, possessing excellent mechanical strength, toughness, wear resistance, and chemical resistance, and are widely used in automotive, electronics, textile, and packaging applications. However, pure PA materials exhibit deficiencies in certain functional scenarios: poor antistatic performance, relatively high friction coefficient, and lack of antibacterial activity, which limit their application expansion in precision electronics, medical, and high-end manufacturing fields.

Nano metal oxide fillers provide an effective pathway for the functionalization of PA. Among numerous nanofillers, copper oxide (CuO) has attracted considerable attention due to its unique advantages: copper is abundant in the Earth's crust and low in cost; CuO possesses good chemical stability and is resistant to further oxidation; nanoscale CuO has high specific surface area and surface activity, imparting antibacterial and antistatic functions to composites. Furthermore, CuO in PA matrices can be partially reduced to metallic copper (Cu) by the weak reducing capability of amide groups, and the in-situ generated Cu nanoparticles possess superior electrical and thermal conductivity.

1.2 Research Status

1.2.1 Preparation Methods for PA6/CuO Composites

The preparation of PA6/CuO composites is mainly divided into two categories: in-situ polymerization and melt blending.

In-situ polymerization/in-situ solution method: Xu et al. developed a novel in-situ solution synthesis strategy using CuO as the metallic copper source, exploiting the weak reducing capability of PA6 terminal amino groups to directly reduce CuO to metallic Cu without additional reducing agents . The key advantages of this method are: the reducing atmosphere of PA6 can directly reduce CuO; the encapsulation effect of PA6 on metallic Cu nanoparticles effectively prevents particle agglomeration and oxidation. TEM analysis revealed that the in-situ generated Cu nanoparticles had sizes of approximately 4–5 nm, uniformly dispersed in the PA6 matrix .

Melt blending method: Zahran et al. prepared Nylon 66/CuO nanocomposites using a melt-mixing process, with CuO contents of 1–4 wt% . This method is simple in process and low in cost, but the dispersion uniformity of nanoparticles depends on mixing conditions and surface treatment. SEM analysis indicated that CuO nanoparticles exhibited homogeneous dispersion in the nylon 66 matrix .

Ultrasound-assisted extrusion method: Sierra-Ávila et al. developed an ultrasound-assisted extrusion method to prepare Nylon 6/Cu nanocomposites, with Cu concentrations ranging from 0.01–0.50 wt% . This method utilizes the cavitation effect of ultrasound to improve nanoparticle dispersion, with optimal dispersion achieved at 0.10% concentration.

Co-precipitation method: Recent research employed a room-temperature reaction of barium hydroxide with copper sulfate pentahydrate in methanol to prepare CuO@$BaSO_4$ co-precipitate filler in one step, with particle sizes of 30–50 nm and high compatibility with the PA6 matrix .

1.2.2 Preparation of PA66/CuO Composites

Research on the preparation of PA66/CuO composites is relatively limited, mainly focusing on melt mixing combined with irradiation modification. Abdeldaym and Elhady synthesized CuO nanoparticles via aqueous precipitation and prepared Nylon 66/CuO nanocomposites with different CuO contents (1–4 wt%) and different gamma irradiation doses (100, 200, 300 kGy) through melt mixing . The role of gamma irradiation lies in inducing polymer chain crosslinking and grafting, producing synergistic effects with CuO nanoparticles.

1.2.3 Reduction and Functionalization Mechanisms of CuO

The behavior of CuO in PA matrices is distinctive. Research has shown that CuO is not reduced when heated with ε-caprolactam at 200°C for 5 hours. However, during PA6 synthesis, terminal amino groups (-NH_2) possess weak reducing capability and can reduce CuO to metallic Cu, while the amino groups are oxidized to oxime groups (=NOH) . This in-situ reduction mechanism enables uniform dispersion of metallic Cu nanoparticles in the PA6 matrix, with PA6 layer encapsulation improving both dispersion and oxidation resistance .

1.3 Research Objectives and Content

This study aims to systematically investigate the enhancement effects and mechanisms of nano-CuO powder on polyamide composite properties. Specific research contents include:

(1) Investigating the influence patterns of CuO content on PA6 composite mechanical properties (tensile strength, elongation at break, impact toughness, elastic modulus);

Chapter 2 Experimental Methods

2.1 Materials

ε-Caprolactam was obtained from BASF Chemical Reagent Company China (Beijing, China). Adipic acid was supplied by Paini Chemical Reagent Factory of Zhengzhou. Formic acid was provided by Tianjin Kemel Chemical Reagent Company Ltd. Copper oxide was supplied by Nanjing Milan Chemical Company Ltd . For PA66 composites, nylon 66 pellets and CuO nanoparticles (synthesized via aqueous precipitation) were used .

2.2 Preparation of PA6/CuO Composites via In-Situ Polymerization

Proper amounts of copper oxide, adipic acid, ε-caprolactam, and distilled water were sequentially added to a 250 mL three-neck flask. The resultant mixed solution was then heated with a stirring constant-temperature electrothermal cover from room temperature to 190°C over a period of over 3 hours under magnetic stirring, followed by heating at 230°C for 0.5 hours to release water vapor. Upon completion of water vapor releasing, the reaction mixture was gradually elevated from 230°C to 260°C within 1.5–2 hours to allow reaction. At the end of the reaction, the pressure of the reaction system was gradually reduced to −0.08 MPa with a vacuum circulating

water pump within about 1.5 hours, thereby affording the target product. Thereafter, crude nano-Cu/PA6 composites were cast molded into final dumbbell-like specimens .

2.3 Preparation of PA66/CuO Composites via Melt Mixing

CuO nanoparticles were synthesized using an aqueous precipitation method. Nylon 66/CuO-based nanocomposites were prepared through a melt-mixing process using CuO nanoparticles with differing contents (1, 2, 3, and 4 wt%) and varying doses of gamma radiation (100, 200, and 300 kGy) .

2.4 Preparation of PA6-CuO@$BaSO_4$ Composites

By simply using barium hydroxide ($Ba(OH)_2 \cdot 8H_2O$) to react with cupric(II) sulfate pentahydrate ($CuSO_4 \cdot 5H_2O$) in methanol under room temperatures, cupric oxide and barium sulfate co-precipitates (CuO@$BaSO_4$) were one-step prepared, and the methanol can be recycled after separating the co-precipitates. The CuO@$BaSO_4$ co-precipitates were then blended with polyamide 6 (PA6) to prepare PA6-CuO@$BaSO_4$ composites and composite fibers via melt extrusion and melt spinning methods .

2.5 Characterization Methods

TEM analysis: A TEM was performed at an accelerating voltage of 200 kV to observe the dispersion behavior of copper nanoparticles in the PA6 matrix; as-prepared nano-Cu/PA6 composite was ultramicrotomed into thin pieces of around 50 nm thick with a Leica EMUC6 for TEM analysis .

XRD analysis: XRD patterns were recorded with a D/max 2550 VX-ray diffractometer (Philips, Holland; Cu Kα radiation, $\lambda = 1.54178$ Å) .

UV-vis absorption spectrum: Measured over a wavelength range of 200–1000 nm with a Cary 100 spectrophotometer; the specimen for UV-vis analysis was obtained by dissolving a proper amount of as-prepared nano-Cu/PA6 composite in formic acid .

Mechanical testing: Tensile strength was evaluated with a WDW-10D testing machine according to Chinese national standard GB/T 1040-1992, and impact strength was evaluated with a ZBC-1400-2 testing machine according to GB/T 1040-1993 .

SEM analysis: Used to examine the morphology of nylon 66/CuO nanocomposites .

Fourier-transform infrared spectroscopy (FTIR): Used to verify structural changes in nylon 66 .

Chapter 3 Results and Discussion

3.1 PA6/CuO Composites via In-Situ Polymerization

3.1.1 Morphology and Dispersion

TEM and EDXS images revealed that CuO was reduced to metallic Cu during the ring-opening polymerization of ε-caprolactam, and the resulting Cu nanoparticles were uniformly dispersed in the PA6 matrix . The reduction of CuO to elemental Cu was facilitated by the weak reduction capability of the PA6 amino groups, and the barrier effect of polyamide 6 to the atmosphere kept the properties of as-fabricated Cu nanoparticles and avoided aggregation .

3.1.2 Mechanical Properties

The tensile strength of the 0.3 wt% sample increased by 47.5% compared to that of nylon 6 . Additionally, the presence of copper nanowires in nylon 6 composites increased crystallinity by 1.38%, decreased crystallization temperature by 1.79°C, and increased tensile strength by 77.4% compared to the pure polymer .

3.1.3 Tribological Performance

The nano-Cu/PA6 composite exhibited improvement in antifriction and antistatic properties, thereby broadening the application of polyamide 6 .

3.2 PA66/CuO Composites with Gamma Irradiation

3.2.1 Structural Changes

The combination of CuO nanoparticles and gamma irradiation caused nylon 66 to undergo structural changes verified through X-ray diffraction measurement and Fourier-transform infrared spectroscopy .

3.2.2 Mechanical Properties

The influence of CuO nanoparticles and gamma irradiation significantly augmented the flexural strength and flexural modulus of the nanocomposites. However, this addition led to a decline of elongation at break . To better understand the tensile mechanism, correlation of tensile strength using theoretical models premised on Money, Einstein, and Pukanszky was undertaken. The optimal deviation was exhibited by the Pukanszky model using tensile plots on an experimental basis .

3.2.3 Electrical Properties

Electrical conductivity increased with CuO nanoparticle content and gamma irradiation. For every sample, the prevailing transport mechanism was the Poole–Frenkel emission. This finding is encouraging for the development of innovative materials with augmented tensile strength and nanoelectronic devices .

3.3 PA6/CuO@$BaSO_4$ Composites

3.3.1 Morphology and Dispersion

The as-prepared CuO@$BaSO_4$ co-precipitates have a regular size of 30–50 nm and are highly compatible with the PA6 matrix. Further investigations disclosed that the CuO@$BaSO_4$ co-precipitates had good dispersion in the PA6 matrix .

3.3.2 Mechanical Properties and Thermal Stability

The CuO@$BaSO_4$ co-precipitates synchronously improved the mechanical properties and thermal stability of PA6 composites .

3.3.3 Antibacterial Performance

The PA6-CuO@$BaSO_4$ composites showed excellent antibacterial performance and high spinnability. The bactericidal rate reached 100% when adding a tiny content (0.24 wt%) of Cu^{2+} into PA6 fibers .

3.4 Nylon 6/Cu Nanocomposites via Ultrasound-Assisted Extrusion

For nylon 6/Cu nanocomposites prepared by ultrasound-assisted extrusion, the presence of the crystalline phase γ was confirmed by X-ray diffraction, and the γ phase showed greater increase than the α phase as a function of copper nanoparticle concentration. This process was attributed

to secondary crystallization. The chemical composition of the nanoparticles was determined to be a blend of metallic copper and cupric oxide. The concentration of 0.10% exhibited the best dispersion. For the 0.10% concentration nanocomposite, higher mechanical properties were obtained: tensile strength increased by 8.9%, and tensile modulus increased by 25.4% .

Chapter 4 Conclusions

4.1 Major Findings

(1) **PA6/CuO composites via in-situ polymerization**: The weak reduction capability of PA6 amino groups enables in-situ reduction of CuO to metallic Cu nanoparticles (4–5 nm) uniformly dispersed in the PA6 matrix. At 0.3 wt% CuO content, tensile strength increases by 47.5%, and copper nanowire-containing composites show 77.4% tensile strength improvement.

(2) **PA66/CuO composites with gamma irradiation**: The combination of CuO nanoparticles and gamma irradiation significantly augments flexural strength and flexural modulus. Electrical conductivity increases with CuO content and irradiation dose, with Poole–Frenkel emission as the prevailing transport mechanism.

(3) **PA6/CuO@$BaSO_4$ composites**: One-step room-temperature co-precipitation produces 30–50 nm CuO@$BaSO_4$ particles with high PA6 compatibility. At only 0.24 wt% Cu^{2+} addition, bactericidal rate reaches 100% while mechanical properties and thermal stability are simultaneously improved.

(4) **Ultrasound-assisted extrusion**: Provides an alternative melt-processing route for Nylon 6/Cu nanocomposites, with 0.10% Cu concentration yielding optimal dispersion and 8.9% tensile strength improvement.

4.2 Innovation Points

(1) Systematic comparison of different preparation strategies (in-situ polymerization, melt mixing, ultrasound-assisted extrusion, co-precipitation) for PA/CuO composites.

(2) Elucidation of the in-situ reduction mechanism of CuO by PA6 amino groups, enabling direct formation of metallic Cu nanoparticles without external reducing agents.

(3) Demonstration of synergistic effects between CuO nanoparticles and gamma irradiation in PA66 composites.

(4) Development of CuO@$BaSO_4$ co-precipitate filler achieving 100% bactericidal rate at ultra-low loading (0.24 wt%).

4.3 Future Work

(1) Scale-up of in-situ polymerization for industrial production of PA/CuO composites.

(2) Long-term stability testing of antibacterial performance under repeated washing and aging conditions.

(3) Investigation of synergistic effects of hybrid fillers (CuO combined with other functional fillers).

(4) Development of multi-scale models to predict mechanical and functional properties of PA/CuO composites.

References

[1] A. Abdeldaym, M.A. Elhady, Role of copper oxide nanoparticles and gamma irradiation in optimising mechanical and the DC-electrical properties of nylon 66, J. Compos. Mater. 54 (2020) 3595–3610.

[2] X. Xu, et al., Fabrication of nano-Cu/PA6 composite by in-situ reduction of CuO during ring-opening polymerization of ε-caprolactam, New J. Chem. (2014).

[3] A. Abdeldaym, M.A. Elhady, Role of copper oxide nanoparticles and gamma irradiation in optimising mechanical and the DC-electrical properties of nylon 66, J. Compos. Mater. 54 (2020) 3595–3610.

[4] B. Xu, C. Yi, Preparation of nano-CuO@$BaSO_4$ under room temperatures and its application in PA6 composites, J. Reinf. Plast. Compos. 44 (2025) 1047–1059.

[5] R. Sierra-Ávila, et al., Synthesis and thermomechanical characterization of nylon 6/Cu nanocomposites produced by an ultrasound-assisted extrusion method, Adv. Mater. Sci. Eng. 2018 (2018) 4792735.

[6] Z. Zhang, et al., Preparation method of nano-copper-polyamide composite material, Chinese Patent CN201310295266.9 (2013).

[7] R. Sierra-Ávila, et al., Synthesis and thermomechanical characterization of nylon 6/Cu nanocomposites produced by an ultrasound-assisted extrusion method, Adv. Mater. Sci. Eng. 2018 (2018) 4792735.

[8] B. Xu, C. Yi, Preparation of nano-CuO@$BaSO_4$ under room temperatures and its application in PA6 composites, J. Reinf. Plast. Compos. 44 (2025) 1047–1059.

[9] R. Sierra-Ávila, et al., Synthesis and thermomechanical characterization of nylon 6/Cu nanocomposites produced by an ultrasound-assisted extrusion method, Adv. Mater. Sci. Eng. 2018 (2018) 4792735.

[10] B. Xu, C. Yi, Preparation of nano-CuO@$BaSO_4$ under room temperatures and its application in PA6 composites, J. Reinf. Plast. Compos. 44 (2025) 1047–1059.